

ROOF SYSTEMS BUILD

An Engineering Design Project

By Conner Glover

Objective: Constructing a functional miniature roof structure on a tissue box model to explore roofing systems, materials, and load distribution.



Brainstorming

According to the criteria, I needed a dormer on my roof, so I chose the Gambrel with dormer roof because the resources are readily available, it's quite simple, and it's the most common design with a dormer. Before beginning the construction, I used trigonometry to understand the dimensions of the roof before cutting and guessing giving me a proper start and eliminating the guesswork. I decided that the angle of the house should be 45 degrees, as I could use a 45-45-90 triangle to simplify the math and it would standardize the dimensions better. I didn't plan out any of the sheathing, plywood, or shingles; I just left it up until the end.



RIGHT

Successfully completed the project. Planned everything out before working. Scanned the house as a 3D Model..

WRONG

Over thought it. I was late on my presentation. Didn't take enough pictures during the process.

LEARNED

I should try not to perfect everything and work in a timely matter.